

Executive Summary

Building a new Large Modular Reactor (LMR) plant in the U.S. requires an integrated programme of site evaluation, licensing, design, construction and commissioning. The project scope spans site characterisation, regulatory approvals (under NRC's 10 CFR Part 52 processes or the new Part 53), engineering design and fabrication of reactor modules, civil works and Balance-of-Plant (BOP) systems, grid connection and long-term operations planning. Key objectives include achieving strict safety and environmental standards, financing the multi-billion-dollar project cost, and meeting schedules for deployment. The regulatory pathway typically involves pre-application engagement with the NRC, then either an Early Site Permit (ESP) or direct Combined License (COL) review (or both) [26†L163-L172]

[1†L47-L54] . Throughout, public and stakeholder involvement (via meetings and hearings) is mandated [26†L56-L64] [26†L103-L110] . In parallel, detailed site studies (geology, hydrology, seismicity, population and land use) and an Environmental Impact Statement under NEPA must be completed [15†L333-L342] [26†L97-L102] . Technical design choices include proven large Light Water Reactors (e.g. AP1000, ESBWR, APR1400) and emerging modular/advanced options (e.g. NuScale SMR, GE BWRX-300, Holtec SMR-160, TerraPower Natrium, X-energy Xe-100) featuring passive safety and innovative coolants [29†L113-L120] [30†L137-L145] . Major tasks then progress through module fabrication, on-site assembly, construction management with strict QA/QC, and finally commissioning tests to verify all safety systems (ITAAC). Long-term operations involve fuel cycles (every 1–2 years for refuelling), maintenance outages, spent fuel cooling/storage, and eventual decommissioning plans. Financing must cover high capital costs (CAPEX often many \$/kW) and ongoing costs (OPEX), using models like regulated rate-base financing, loan guarantees or power-purchase agreements. Risk management (cost/schedule overruns, insurance under the Price-Anderson Act) and contingency budgets are essential. A typical schedule from project start through operation is on the order of a decade, with NRC licensing and construction as the critical path. Tables below compare reactor designs, regulatory steps and project phases; and diagrams illustrate a project timeline and a high-level plant architecture.

Project Scope and Objectives

The project envisages constructing a modular nuclear plant (LMR) comprising one or more nuclear reactor modules (each typically 50–500 MWe) and associated BOP systems (steam generators, turbine/generators, cooling infrastructure, etc). Objectives include: low-carbon baseload generation (replacing fossil plants), scalability via factory-built modules, high reliability, and enhanced safety through modern design. This involves detailed pre-feasibility and site studies, design of the reactor and power island, compliance with NRC and environmental regulations, community outreach, arranging financing, and ensuring a trained workforce. The project must deliver a final product (typically $\sim 10^3$ MWe total) that meets regulatory criteria for safety (no undue risk) and environmental impact (minimal emissions) while remaining cost-competitive over its lifetime. According to DOE, advanced SMRs (up to hundreds of MWe) offer small footprints, lower upfront CAPEX and modular scalability [17†L25-L28] – advantages the project aims to capitalise on.

Regulatory and Licensing Pathway

Licensing follows the NRC's 10 CFR Part 52 process, optionally augmented by the new 10 CFR Part 53 for advanced reactors. Key NRC approvals include:

- **Pre-application Engagement:** Early meetings with NRC staff (and DOE if applicable) to identify major issues. This phase is flexible in duration and helps refine the safety analysis before formal submittal.
- **Early Site Permit (ESP, optional):** NRC review of site suitability (safety, environmental, emergency planning) independent of any specific reactor design [10†L35-L43] [26†L163-L172] . An ESP application covers site boundaries; seismic/meteorology/hydrology data; projected population and exclusion/evacuation zones; alternative sites evaluated; proposed location and number of reactors; cooling system; accident dose analyses; and emergency plans [26†L169-L177] . NRC review (including Draft/Final EIS) typically takes ~1–3 years. The ESP, once issued, is valid 10–20 years (renewable) [26†L191-L196] .
- **Design Certification (DC):** NRC rulemaking to certify a reactor design's safety (40-year term, renewable) [4†L35-L44] . Several designs (e.g. Westinghouse AP1000, GE ESBWR, APR1400, NuScale US-600) are already certified [4†L62-L72] [30†L137-L145] . If using a certified design, the COL can reference it; otherwise a site-specific design review under Part 52 applies. DC review for a new design typically takes 3–5 years.
- **Combined License (COL):** NRC review of the site-specific COL application (10 CFR 52) which, upon issuance, authorises construction (and, after meeting ITAAC, operation) [26†L137-L146] . The COL must include a Final Safety Analysis Report (FSAR) with all technical, site, and operational programs. If no ESP or DC is referenced, NRC reviews site safety and design together as in a two-step (Part 50) process [26†L137-L146] . NRC also produces a Draft EIS (for comment) and Final EIS (NEPA compliance) [26†L97-L102] . The Atomic Energy Act mandates a mandatory hearing (Licensing Board) on the COL [26†L103-L110] . NRC aims for ~2–3 years of formal review for a high-quality COL application (expedited cases, e.g. X-energy, have aimed for ~18 months vs. ~36 months normally [37†L90-L94]). Once a COL is issued, construction can start; operation only after all Inspections, Tests, Analyses and Acceptance Criteria (ITAAC) are fulfilled [26†L152-L159] .

The NRC (Office of Nuclear Reactor Regulation) is the lead federal agency for these steps, working under the Atomic Energy Act and in compliance with NEPA. Table 1 (below) summarises major licensing steps, responsible agencies and typical durations. Other required approvals include DOE for loan guarantees (if used), state utility commissions (for regulated-cost recovery), and local permits (land use, building codes). EPA or state environmental agencies will participate as NEPA cooperating agencies.

Site Selection and Environmental Review

Site criteria are guided by 10 CFR Part 100 (Reactor Site Criteria) and related Reg Guides. Key factors are: geology/seismic (stable bedrock, no active faults, low SSE hazard); hydrology and flooding (above flood plains, safe cooling-water access, no undue contamination risk) [15†L333-L342] [15†L349-L354] ; meteorology (favourable conditions to disperse heat and any effluents); and **population distribution**. NRC regulations require defining an **exclusion area** (no permanent residences or water intakes) and a **low-population zone** such that worst-case plume doses at the zone boundary do not exceed 25 rem whole-body (or 300 rem thyroid) [15†L374-L382] . A **population center distance** ($\geq 4/3 \times \text{LPZ radius}$) must separate major population centres from the reactor [15†L374-L382] [15†L380-L388] . In practice, sites with generally low surrounding population density are preferred, though higher-density sites may be accepted if overall risk is shown acceptable [13†L79-L88] [13†L90-L100] . Practically,

preferred sites have adequate buffer zones, good transport access (rail/road), and critical services (water, electricity, emergency response) while minimising impacts on environmentally sensitive areas.

Environmental review (NEPA) is integrated into NRC licensing. NRC (as lead agency) will conduct an Environmental Impact Statement (EIS) covering land use, water use, ecosystems, historical/cultural impacts, etc. This involves scoping, publication of Draft EIS for public comment, and Final EIS addressing comments [26†L97-L102] . The Record of Decision (ROD) issues after the Final EIS, determining conditions on construction (for example, protected wetlands, mitigation plans). Any DOE involvement (e.g. on federal land or DOE funding) requires DOE to also comply with NEPA.

Permitting and Public Engagement: In addition to the NRC license, the project will need federal/state permits: e.g. Clean Water Act (water discharge/withdrawal permits from EPA/Corps), Clean Air Act permits (for backup diesel generators), hazardous materials/waste approvals, local building and fire codes, and possibly state-level siting certification (where applicable). Public engagement is mandatory: NRC holds informational meetings on the application [26†L56-L64] and public hearings (intervenor process) during ESP, DC or COL reviews. Citizens and agencies can petition to intervene in the COL hearing [26†L103-L110] . Pro-active stakeholder consultations (town halls, Open Houses) during site selection and EIS scoping are standard practice to address concerns on safety, economic impact, and environmental justice.

Technical Design Choices and Vendor Options

Reactor Types: The project can choose from conventional Light-Water Reactors (LWRs) or advanced modular concepts. Proven certified LWR designs include large PWRs (e.g. Westinghouse AP1000, Korean APR-1400) and BWRs (GE ABWR, ESBWR) [29†L113-L120] [30†L131-L134] . These use uranium oxide fuel and either pressurised (PWR) or boiling (BWR) water cooling. All certified designs incorporate robust safety systems; modern ones (AP1000, ESBWR) use *passive safety* (gravity, natural circulation) for emergency cooling [29†L113-L120] [29†L121-L129] . For a **modular plant**, one might select a smaller “standard” design (e.g. NuScale’s 50 MWe integral PWR modules [30†L137-L145]) or a mid-size LWR (e.g. 300–600 MWe modules like the planned Westinghouse AP300 or GE BWRX-300). Alternatively, **advanced reactors** use non-water coolants: e.g. TerraPower’s *Natrium* (345 MWe sodium-cooled fast reactor with molten-salt thermal storage) [30†L150-L158] , X-energy’s *Xe-100* (pebble-bed high-temperature gas reactor, 320 MWe total), and various molten-salt or lead-cooled designs in development. These advanced concepts promise inherent safety features (e.g. negative temperature coefficient, passive heat sinks, or decay heat removal by physics) that reduce reliance on active pumps [30†L150-L158] [37†L135-L144] .

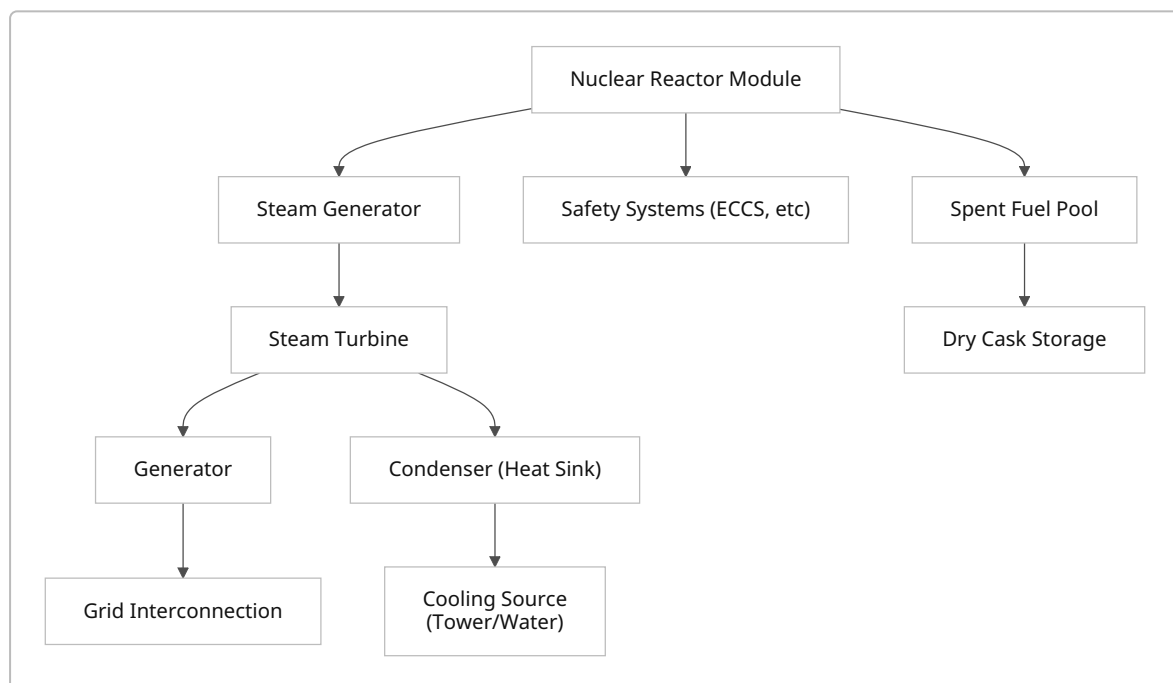
Table 1 compares representative designs (type, output, safety features). Certified designs include ABWR (GE, 1300 MWe BWR), AP1000 (Westinghouse, 1150 MWe PWR), ESBWR (GE, 1500 MWe BWR), APR-1400 (KEPCO, 1300 MWe PWR) and NuScale US-600 (NuScale, 12×50 MWe PWR) [29†L113-L120] [30†L137-L145] . SMR/MRs under NRC review or in pre-application include GE’s BWRX-300 (300 MWe, passive BWR), Westinghouse AP300 (300 MWe PWR, passive), Holtec SMR-160 (160 MWe PWR, passive), TerraPower Natrium (345+ MWe SFR with storage), X-energy Xe-100 (320 MWe HTGR), and others [30†L150-L158] .

Safety Systems: All designs must meet NRC Safety Criteria (10 CFR 50 Appendix A or Part 53 performance criteria). Passive features (gravity-driven cooling, natural circulation) are emphasised. Redundant diesel generators or gas turbines provide back-up power; containment structures must withstand internal pressures and external hazards. Physical security (co-located armed response), seismic qualification, and aircraft-impact assessments are also required for modern designs. Many new

designs simplify the safety case by reducing system complexity (e.g. integral PWR loops in NuScale mean fewer large components) and using robust materials (e.g. high-temperature ceramics).

Balance of Plant (BOP): This includes the steam turbine/generator, condensers, feedwater and auxiliary systems, electrical switchyard, and non-nuclear buildings (control, administration, maintenance). The reactor's secondary side (steam cycle) is largely conventional. The design must consider dry or wet cooling (towers vs. river/sea water) depending on site water availability. Heat rejection systems (cooling towers or once-through flow) must meet environmental limits (thermal discharge, drift) and any local water-rights laws. The plant may include services for future cogeneration (steam for industrial use or district heating) if planned.

System Architecture: High-level architecture is illustrated below. Nuclear modules (the “nuclear island”) produce heat, driving a steam generator and turbine (the “turbine island”) which feeds power to the grid. Cooling is provided via a condenser and tower/water source. On-site spent fuel storage (pool or dry cask) is usually adjacent to the reactor. Passive/active safety subsystems connect to the reactor for emergency heat removal.



Civil Works, Cooling and Grid Interconnection

Civil Works: Site preparation includes grading, foundations, drainage and utilities. The reactor modules and BOP equipment (turbine hall, cooling towers, switchyard) require significant civil structures of reinforced concrete. Containment basemat and building bear seismic loads per criteria. Modular construction reduces on-site work: many major components (reactor vessel, steam generators, turbine sections) are fabricated off-site under QA programs and delivered by special transport. Construction sequencing is critical (see schedule), with close NRC inspection of foundations and embedment of critical support structures. Quality Assurance must follow 10 CFR 50 App. B (QA criteria) throughout procurement and erection.

Cooling Options: Depending on climate and site, cooling may be once-through (river or ocean water, requiring CWA permits) or cooling towers (wet-dry or dry towers, which use more land and capital).

Hybrid systems (partial tower + water) are also possible. Drought and environmental considerations (fish impact, thermal plume) strongly influence choice. For an inland site, a forced-draft tower with wastewater treatment is typical. The design must meet state water use and thermal limits, and include closed-circuit cooling if mandated.

Grid Interconnection: The plant must interconnect to the grid with a suitable substation and transmission lines. This involves a Generator Interconnection Agreement under FERC tariffs, typically requiring feasibility, system impact and facilities studies (which can take ~1–2 years). The project may need transmission upgrades (new lines, transformers) if local grid capacity is limited. Under FERC's recent Order 2023 reforms, interconnection processes are being streamlined, but realistically a multi-year timeline must be expected. Early coordination with the regional transmission operator (RTO/ISO) and local utility is essential. The switchyard and step-up transformers are part of civil scope and must comply with NERC reliability and local regulations.

Construction Management, Supply Chain and Workforce

Construction Management: Due to nuclear complexity, an integrated project team is required. The Owner/Licensee (often a utility or consortium) hires an architect-engineer and construction manager (often the vendor) to oversee contractors. Staged contracting is common: e.g. EPC (Engineering, Procurement, Construction) or separate contracts for nuclear vs. conventional islands. The project follows a strict QA plan (10 CFR 50 App. B/C), with all suppliers qualified and welds/materials tested (ASME section III, etc). International vendors (e.g. turbine supplier) must meet US QA. A construction quality assurance unit monitors all activities and maintains as-built records.

Supply Chain & Modular Fabrication: The modular approach relies on factory fabrication of many components. For example, reactor modules (like NuScale's integrated vessels) or large piping modules are built in controlled environments. Critical equipment (reactor internals, control rods, steam generators, turbine rotors) have long lead times (often 2–3 years) and require nuclear-grade manufacturing. The supply chain must be certified under nuclear QA, and security protocols for sensitive components. Vendor inventories of specialty steel, forgings and instrumentation must be secured early. The construction plan should include rigorous logistics for delivering heavy components (often via barge or rail) to the remote site.

QA/QC and Regulatory Oversight: All work must comply with NRC regulations (10 CFR 50 App. B on QA, 10 CFR 52 ITAAC). Inspections are frequent, and NRC resident inspectors may be assigned early. Third-party inspections (NRC, ASME, owner) check critical welds, concrete tests, and equipment installation. Any non-conformance must be corrected before proceeding. Nuclear plants historically have tight QA (e.g. turbine-build for nuclear must use quality metal), which adds schedule risk if problems occur. A detailed QA manual and training program for all staff is mandatory.

Workforce Needs: Building a nuclear plant requires thousands of skilled tradesmen (pipefitters, welders, electricians) plus engineers and managers. As an example, recent nuclear builds peaked at 3–5,000 workers. A modular project may compress field labour due to factory work, but still needs specialized crews for nuclear installation. Recruitment and training (often via apprenticeship) must align with the schedule. Coordination with local labour unions and vocational schools can help meet workforce demand. Owner must plan for a stable workforce over the ~5-year construction period to avoid delays.

Commissioning, Testing, and Initial Operations

After mechanical completion, commissioning follows a precise sequence under NRC supervision. **Pre-operational Testing** includes cold hydro and pressure tests of systems, followed by hot functional tests without fuel (steam runs, pump heat-up). Each system (reactor coolant, decay heat removal, emergency core cooling, control systems) is tested to demonstrate acceptance criteria. Results are documented for NRC.

Fuel Loading & First Criticality: The licensee loads enriched uranium fuel assemblies (procured from fuel suppliers). For LWRs this occurs after initial tests; advanced reactors may use TRISO or metal fuels (e.g. HALEU). The first criticality (bringing reactor to self-sustained power) is another key milestone, performed in coordination with NRC staff. After reaching power, a power ascension program tests the reactor at increasing power levels, verifying all safety limits.

Initial Operations: Once at nominal power, the plant operates under NRC oversight. Typically a 40-year operating license is issued (plus 20-year renewal possible) [1†L35-L44] . Refuelling is planned (e.g. every 18–24 months for a PWR unit; more frequently for smaller cores). The initial few fuel cycles may involve surveillance samples and performance calibration. Operating staff must be fully trained and licensed (NRC requires a plant-specific training program and operator licensing exams). Early operations often run at reduced power factors until all operational experience and data are confirmed.

Long-Term Operations, Maintenance, Decommissioning and Waste Management

Operations & Maintenance: Over its life, the plant will have periodic refuelling outages, typically ~3–6 weeks annually for SMRs or longer for larger cores. Maintenance plans include major overhaul schedules for turbines and replacement of certain components (like steam generators after ~40 years if needed). An on-site maintenance and spare parts inventory is maintained. Nuclear plants require a safety culture and performance monitoring; NRC inspections and performance indicators (unplanned scrams, safety system availability) are tracked continuously.

Decommissioning Planning: From the outset, NRC requires funding and planning for eventual decommissioning. A Decommissioning Trust Fund (collected over operating life) is held to cover future dismantling costs. The licensee must submit a Post-Shutdown Decommissioning Activities Report (PSDAR) at least 2 years before shutdown, and secure decommissioning funds (as per 10 CFR 50.82). Decommissioning strategies (immediate vs. SAFSTOR) are evaluated, but typically not carried out until end of license.

Waste Management: Spent nuclear fuel is initially stored in a protected spent fuel pool on-site. As the pool nears capacity (typically after ~5 years cooling), older fuel is moved to dry cask storage (10 CFR 72). Most U.S. plants (likely including this one) will store spent fuel on-site indefinitely, since a permanent repository is not currently available. Dry cask systems (licensed under 10 CFR 72) provide robust interim storage. Low-level radioactive waste (radwaste from plant systems) is packaged and sent to approved disposal sites (e.g. Washington State, Utah). The plant must have plans for radwaste processing, segregation and shipment.

Cost Estimates and Financing Models

Nuclear plants have high upfront capital costs. **Capital Expenditure (CAPEX):** Includes site work, reactor modules, BOP, grid interconnection, engineering, and financing costs during construction. Historic U.S. projects ran several thousand \$/kW. International data (OECD) show overnight costs ranging ~\$2,000–7,000 per kW 【41†L25-L29】 (e.g. ~\$3,850/kW OECD median in 2009). Large Gen III reactors can exceed \$4,000–5,000/kW outturn (e.g. some EPR projects 【41†L25-L29】). Modular designs aim to reduce CAPEX via series manufacturing. For a 500 MWe LMR, rough CAPEX might be on the order of \$2–4 billion, excluding financing (or \$4,000–8,000/kW). Cost escalation and financing costs (during the ~5-year build) can easily double the “overnight” CAPEX.

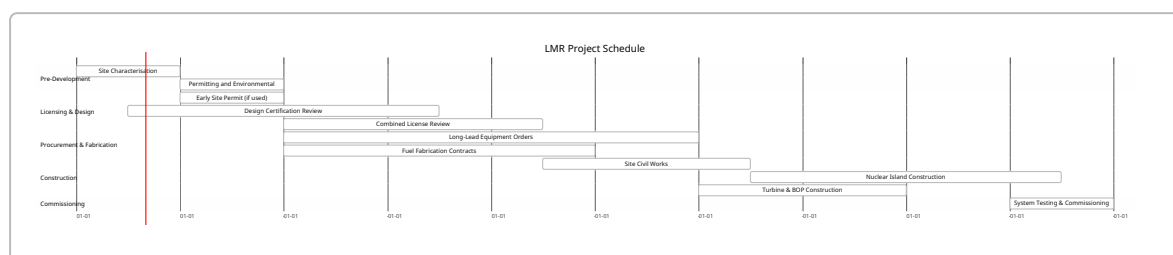
Operating Expenditure (OPEX): Once built, nuclear O&M costs (fuel, staffing, maintenance) are moderate. Fuel costs for LWRs are typically \$0.5–1.0 per MMBtu of heat or ~\$5–10/MWh of electricity (uranium + conversion/enrichment/fabrication). O&M (excluding fuel) runs on the order of \$20–30/MWh. Overall generating cost (after start-up) is often cited as \$40–60/MWh depending on financing. These values depend on capacity factor (assumed ~90%) and long plant life. Including waste/disposal reserves, decommissioning funding and interest, the **Levelised Cost of Electricity (LCOE)** for new nuclear (in OECD studies) has ranged roughly \$50–120/MWh (3% discount, 85% CF) 【41†L25-L29】 . For example, one OECD report gave \$27–102/MWh for various countries at 85% CF 【41†L25-L29】 .

Financing Models: Utilities may finance nuclear via regulated rate-base (recovering costs from customers over time) or via merchant strategies (power sale contracts, equity). U.S. federal loan guarantees (DOE’s Title XVII programme) are available to reduce borrowing costs. Other models include limited recourse project finance (rare for nuclear) or public-private partnerships. Tax credits (e.g. the 2022 Investment Tax Credit for new nuclear) can offset CAPEX. Risk-sharing mechanisms (e.g. guaranteed fixed price contracts, government warranties) have been proposed to help manage cost overrun risk.

Risk Analysis and Contingency: Major risks are construction delays and cost overruns. Budgets must include contingencies (often 20–30% of base CAPEX) for unforeseen issues. Supply chain disruptions (e.g. quality rejects) and regulatory changes are additional risks. The plant will need comprehensive insurance: commercial insurance covers property damage onsite, and *Price-Anderson Act* provides up to ~\$16 billion of off-site liability coverage for public claims 【45†L18-L25】 . Owners pay modest premiums (about \$1.1 million/year per reactor) for the first-tier \$500 M liability 【45†L45-L49】 . The remainder of any nuclear accident damage is covered by assessments on all reactors, backed by Congress if needed 【45†L18-L25】 . Contingency planning also includes emergency response preparedness (onsite and with local/state authorities).

Schedule and Milestones

Mermaid Diagram: Project Timeline



The **critical path** runs through licensing and construction. NRC reviews (ESP/DC/COL) can take ~3–5 years [26†L163-L172] [37†L90-L94] , after which a multi-year construction (3–5 years) occurs. Table 2 below outlines phase-wise deliverables. Early activities overlap (e.g. starting engineering and procurement during NRC review).

Phase	Key Deliverables
1. <i>Site & Concept</i> (Year 0–1)	Site Characterisation Report (geology, hydrology, population); Feasibility Study; Preliminary SAR
2. <i>Licensing</i> (Year 1–4)	ESP Application/Permit; Design Certification Petition (if needed); Final Safety Analysis Report; Draft/Final EIS; COL Application; NRC Hearings; NRC Licenses (ESP/COL)
3. <i>Detailed Design/Procurement</i> (Yr 3–6)	Engineering design packages; Procurement contracts for reactor modules, turbines, etc; Quality plans; Fabrication progress reports
4. <i>Construction</i> (Yr 5–10)	Construction Progress Reports; Completed major structures (Containment, Turbine Hall); ITAAC execution logs; NRC inspections; Equipment installation records
5. <i>Commissioning</i> (Yr 10–11)	Completed pre-operational tests; ASME code tests passed; Fuel load approval; Final Safety Evaluation; Commercial Operation Date (COD)

Critical durations: Historically, a Part 50 two-step licence took ~3–6 years (as for recent SMRs) [37†L90-L94] . Part 52 COL reviews are often ~3 years; designing may run concurrently. Construction of multi-module plants often spans ~4–6 years. The schedule above assumes about 10–11 years from project start to first power, consistent with current expectations for new nuclear builds.

Milestones: Important milestones include NRC acceptance of COL application, NRC issuance of COL, groundbreaking ceremony (often tied to COL), final component delivery, first concrete pour (nuclear island), reactor first criticality, and initial electricity generation. NRC must verify all ITAAC before operation; then the plant receives authorization for fuel load and grid connection. All milestones require close documentation and, at key points, public announcements and regulatory filings.

Tables: Table 3 compares design vendors; Table 4 summarises regulatory/licensing steps with agencies and durations as above.

Each phase’s deliverables (design documents, permits, test certificates) must be completed before moving to the next phase. Frequent project reviews and contingency reserves help manage risks. According to NRC policy, no construction of safety-related structures may occur before key NRC findings (e.g. site suitability and design compliance) are confirmed [26†L137-L146] .

System Architecture Diagram: The block diagram above illustrates the high-level plant architecture: nuclear reactor modules produce steam for turbines, with cooling systems and off-site grid tie-in. Passive safety and fuel storage are integral.

Overall, rigorous adherence to regulatory requirements (with NRC as lead), thorough technical design, and meticulous project execution are essential. This report has outlined each step with references to official guidance and industry experience, to provide a comprehensive roadmap for building an LMR power station from the ground up.

Sources: U.S. NRC regulations and fact sheets [26†L163-L172] [26†L97-L102] [26†L103-L110] [1†L47-L54] , DOE/IAEA nuclear summaries [17†L25-L28] [30†L150-L158] , and industry analyses [37†L90-L94] [45†L18-L25] were used to compile this report. These sources cover licensing rules (10 CFR 50/52/53), site criteria, design data, and economic factors as cited above.
